NG5001 Academic Communication for Graduate Researchers

Unit One

Reading Critically

First Semester

2025-2026

Learning Outcomes

This unit explores the key question of what it means to read critically as a graduate student, particularly in cross-disciplinary contexts. We will read academic texts critically to gain an awareness of how writers across disciplines use certain, shared language conventions to produce critical writing. We will focus on evaluating arguments and explore what a critical literature review is.

Students will:

- Consider the relevance and challenges of cross-disciplinary communication in relation to their own research goals
- To understand what it means to read critically in an academic context
- Evaluate arguments in academic texts across disciplines.
- Consider what makes a literature review critical

1.1 READING CRITICALLY I
(Independent E-Learning)

Communication is contextual. It depends on what we-- as writers, readers, speakers, listeners—bring to the table in terms of perspectives, values, beliefs and ideas. It is therefore important that we understand and recognize the influence of one's own and others' perspectives, values, beliefs and ideas on the ways we communicate.

Critical reading is the first step in developing this awareness. In every discipline, students dedicate substantial time to reading, in order to grasp disciplinary concepts and explore the multiple perspectives, assumptions, and controversies involved. *Critical* reading is essential to appreciating the nuances, contexts, and complexities that underpin these academic discussions.

For graduate students, reading is not only a means of acquiring new ideas, models, and theories that will shape their own research, but also a way to actively engage with information in order *to produce new knowledge*. It is therefore important that graduate students understand how to approach texts with a critical eye.

Reflect:

How many elements of critical reading in the list below do you already employ when you read academic literature? The more 'ticks' you have, the further you have already progressed in becoming a critical reader. Look back at any items that you aren't already doing. Consider how you might incorporate these elements of critical reading into your habitual approach to study.

When I read an academic text I:

- try to work out what the authors are aiming to achieve;
- try to work out the structure of the argument;
- try to identify the main claims made;
- adopt a sceptical stance towards the authors' claims, checking that they are supported by appropriate evidence;
- assess the backing for any generalizations made;
- check how the authors define their key terms and whether they are consistent in using them;
- consider what underlying values may be guiding the authors and influencing their claims;
- keep an open mind, willing to be convinced;
- look out for instances of irrelevant or distracting material, and for the absence of necessary material;
- identify any literature sources to which the authors refer, that I may need to follow up.

What does it mean to be critical?

A critical approach to research in academic contexts involves employing a friendly scepticism (Wallace & Wray, 2021) when reading academic material. This means that we are open-minded, but analytical, asking questions that allow us to deconstruct arguments and concepts, in order to look beneath the surface of a text. For example, we may be looking for the author's real purpose, or a hidden agenda, or we might 'test' the author's claims against what we know about the world. Authors rarely state everything explicitly, and often make assumptions about their audience (background knowledge, values, and attitudes), which affects how they write. Therefore, the critical reader must be aware of the implicit messages. Critical reading also involves evaluating the accuracy of an author's statements and the soundness of an author's reasoning that leads to conclusions.

In academia, one of the most common questions asked by critical readers is 'Have you given me sufficient grounds for accepting your claim?' Asking this question need not imply suspecting the author is untruthful. It is usually not a matter of truth, but of viewpoints, interpretation and significance. The critical reading of a text is rarely about questioning the facts. Mostly it is about assessing the quality of the case that has been made for interpreting and evaluating the facts. Thus, the critical reader is interested in whether there is sufficient evidence to support a claim, whether there is another possible interpretation that has not been considered, and perhaps whether the authors have argued convincingly that their interpretation applies to other cases. We think about the extent to which the claims and supporting evidence in a text – which satisfied the authors – also satisfy us.

One can be a critical reader by focusing on several potential objects of scrutiny, including:

- Any evidence provided in the account, and whether it is adequate to support the author's claims;
- Whether the reasoning of the author's argument follows logically to the conclusion that has been drawn;
- Explicit or implicit indications of the author's values and assumptions;
- The match between this author's claims and those of other authors; and
- The match between the author's claims or predictions and the reader's own research evidence or knowledge.

Critical reading = critical writing

Your critical reading of others' work will usually be in preparation for producing your own written text. In other words, critical reading helps you become a critical writer. This marriage of reading and writing has many benefits. First, you will develop a sense of what is and is not a robust piece of research – essential when you come to plan your own. Second, you will soon begin identifying limitations in existing research knowledge that your investigation can address. Third, the attention you pay to different authors' texts will naturally affect the quality of your own writing. You will soon:

- demand of yourself evidence to back up your claims;
- be alert to the possibility of making an illogical jump in your reasoning;
- become sensitive to your own assumptions and how they might affect your claims; and
- realize the importance of checking the literature thoroughly to ensure that your understanding is sufficiently deep.

In short, you will develop a mature academic style of writing that is both fair and discerning in its accounts of others' work, and that maximizes the opportunity for others to take seriously what you have to say.

Task 1: Reading and Viewing

(a) View E-Lecture 2A and E-Lecture 2B (both found in Canvas, Unit 1) to learn how to define critical reading and what makes a literature review critical.

(b) Read Chapter Four of Wallace and Wray (2021) to learn how to ask critical questions of a text. Focus your reading on pp. 36-43.

Criticality and cross-disciplinary communication

Critical thinking and critical reading involve analytical skills that transcend specific disciplines. In fact, developing critical thinking and reading skills necessitates using these skills in many different contexts (Facione, 1990, p. 5).[1] Cross-disciplinary communication (sharing your research with those from various other disciplines) therefore provides multiple different contexts for scholars to develop critical thinking, critical reading and critical writing skills.

An important goal of the academic is to increase research impact and influence the wider community and various decision-makers therein, many of whom are non-disciplinary specialists. This also includes, increasingly, communicating to the general public, which has become an important feature for legitimating various disciplinary activities (in science and engineering, for example).

Cross-disciplinary communication is more important today than ever especially in the context of our increasingly complex world. Many real-world problems today (climate change, for instance) cut across traditional categories of knowledge, and bring together specialists from various disciplines to develop solutions that are broader and more comprehensive than one specialization. If specialists are unable to effectively communicate across disciplines and advocate their position, it is likely that old paradigms of knowledge (and destructive patterns that derive from them) will remain unchallenged.

In this course, through all the units, we will attend to the ways critical thinking and communication language skills, support effective cross-disciplinary communication.

Reflect:
Writing in your discipline, what are some ways you discount or ignore other perspectives on issues?
How might communicating to others from other disciplines improve your approach to various issues in your research?

How to read critically

In the rest of unit 1.1, we will look at how you can identify authors' arguments and judge the adequacy of the backing they offer for their claims.

Evaluating arguments

Critical reading often begins with identifying and evaluating the arguments in a text. An argument consists of a **conclusion** and its **warranting**. The conclusion (which comprises one or more claims that something is, or should be, the case) is only half of

[1] Facione, P. A. (1990). Critical Thinking: A Statement of Expert Consensus for Purposes of Educational Assessment and Instruction. Millbrae, California, USA: The California Academic Press.

an argument. You can legitimately ask of any set of claims: ‘Why should I believe this?’ Thus, the other half of the argument is the warranting, which is the justification for why the claim(s) in the conclusion should be accepted. This includes the reason and evidence for accepting the conclusion. The evidence is likely to be based on the authors’ research or professional experience, or else it will draw on others’ evidence, as reported in the literature. This conception of ‘argument’ can be applied to single sentences, paragraphs, chapters, even entire dissertations or books. It can be used to identify and evaluate what is said in the texts you read, and also to ensure your own scholarly writing is well constructed.

The next step in critical reading is to evaluate whether the warranting provided for a claim is adequate to make the claim convincing. The claims in the conclusion need adequate warranting for an argument to be convincing. Warranting is adequate when you, as the reader, are satisfied both that there is sufficient evidence and that this evidence is of an appropriate kind. The reader might feel that a single voice does not carry much weight and so look for other, supporting evidence. However, the reader might equally decide that the point of the claim is not that it is necessarily a majority view, but that it exists at all. In such a case, the reader might be satisfied that even if this view is restricted to one person, it is sufficient for warranting the claim.

Note that people may differ in their views about what counts as adequate evidence. This is because the strength of the warranting depends only indirectly on the evidence itself. The relationship is mediated by our interpretation. The reason why critical readers in one discipline might question the adequacy of the warranting of claims forming the conclusion of a research paper in another discipline is usually because they differ from the author in their judgement about the amount and quality of evidence necessary for warranting the acceptance of that conclusion. Such views about adequacy cannot be easily generalized. They will often need to take into account the nature and purpose of the study and also the reader’s other knowledge, experience, and interests in reading the text. This is especially important in cross-disciplinary contexts.

In all cases, a claim is as convincing as the adequacy of the warranting that justifies it. Whether you are writing about your own research or commenting on someone else’s, you need to warrant your claims. So, when you are commenting on what others have claimed about their work, you must be careful that your counter-claims are warranted. It is rather easy to criticize the shortcomings of others’ conclusions, and then to draw similarly flawed conclusions oneself! You should demand a convincing warranting for every conclusion that you read about. Also, you should demand of yourself that every conclusion you draw is adequately warranted.

How to ask critical questions of a text

- What are the writers trying to achieve in writing this? (e.g. how might they trying to persuade you or influence your point of view?)
- What are the author's credentials? (e.g. area of expertise; affiliations, etc.)
- What are they claiming?
- How convincing are their claims?
    - Are the claims supported by evidence? What kind of evidence?
    - How adequate is the warranting?
    - How adequately are claims supported by theoretical orientation?
    - Can you identify any implicit values, and to what extent does any value stance affect claims?
    - To what extent are claims supported or challenged by others' work?
    - To what extent are claims consistent with my experience?

Task 2: Online Discussion

Read the following excerpt from Thompson (2010) and then complete Part 1 and Part 2.

Part 1: Connections

Apply what you have learnt about critical reading in this Unit. Post responses to your group's forum in Canvas Discussions, answering the following questions:

a) How convincing are the author's claims, and why?
In your answer, you can briefly evaluate the strength of the evidence and reasoning provided, and/or, identify any key assumptions or weaknesses that affect the persuasiveness of the argument.

b) In what ways do you think the author's ideas could be applied to your specific discipline? Choose one idea and explain how it might relate to your field or your experiences within it.

Your answer to the questions above should be about 100-150 words each.

Part 2: Responding to peers

Read some of your peers' responses. Reply briefly to at least **one** post from your classmates in this discussion. In your reply, include:

- At least one comment to their post
- At least one question about their post

Make sure your response is thoughtful, respectful, and adds something constructive to the conversation.

In the following excerpt, Thompson (2010) explores the contexts and intellectual consequences of the adoption of cross-disciplinary methods.

The Disciplines and Disciplinarity

Let us begin with the disciplines. There are several aspects of disciplines that are thought to typify them and provide them with their intellectual strength. These aspects are central to the popular beliefs about the disciplines. And they provide a framework for the defence of the integrity of disciplines against attempts to loosen the grip on disciplinary activity with the advance of several forms of what is called 'cross-disciplinarity', which are discussed in a moment.

A1 The first characteristic argued to typify traditional disciplines is that they display a certain unity: a unity in their problematics, in their categories, and in their techniques of investigation. This unity provides the disciplines with a coherent intellectual field or purpose, associated with a singular and homogeneous 'object' of investigation: the biological world, the economy, medical conditions, etc.

A2 The second characteristic aspect of disciplines is a certain rigour in respect to their procedures and methods of investigation. Such a rigorous discipline ensures against the perils of sloppy thinking, against the dilution of their approaches which threatens to undermine the forcefulness, authenticity and authority of their analytical results and truth claims.

A3 The third aspect is a certain autonomy in their modes of existence. This autonomy has two basic features. They are autonomous from each other, and they are autonomous from the powers or authorities that might wish to appropriate them. There is all manner of institutionalization of the disciplines designed to ensure against possible encroachments on their separated territorial identities: professionalization, specialist organizations of support and dissemination, academic departmentalization, publishing outlets, etc.

Role of aspects

What these three basic characteristic aspects do is to enable the disciplines to maintain their independence, from one another and from those who might wish to control them. They support the self-confidence of the disciplines, providing them with a seeming strength, authority and singularity of purpose. They enable 'boundary maintenance' (Abbot 2001), which is such an important part of the 'disciplinary complex'. The disciplines fiercely defend their particular intellectual and organizational 'patch'.

Counter claim (Weak claim.

If this represents in outline the popular beliefs about the disciplines what are we to make of these 'defensive' claims? The argument made here is to suggest that the disciplines display few if any of these features in their actual day to day practices: they are not unified, they are not rigorous, and nor are they autonomous in the senses outlined above. In fact the disciplines are always already compromised, they are always subject to cross-fertilizations, and they are riddled with disputes.

Take the discipline of geography as an example. Is this unified? Hardly. It is riddled with disputes, for instance between 'physical geography' and various forms of 'cultural geography'. Similarly with psychology: here one of the main lines of fracture is between 'behavioural' and 'social' psychologists. These hardly speak to one another. But I would suggest that these are not isolated instances, and that such disputes – which involve issues associated with all three of the central aspects of disciplinarity outlined above – are an enduring feature of all the disciplines, at least to some extent. Obviously some display more of the characteristics of unity, rigour and autonomy than others, but few are completely exempt from some compromise and accommodation in respect to these.
[…]

Why the Current Emphasis on Cross-Disciplinarity?

Whatever one chooses to term the current enthusiasm for non-single disciplinary approaches, whether it be interdisciplinarity, transdisciplinarity, or cross-disciplinarity as favoured here, the underlying issue is the same. These are promoted and celebrated as offering the only appropriate approaches to current problems and concerns, basically because of the perceived 'complexity' of the modern world. Thus a single disciplinary approach is not considered adequate to analyse or describe 'globalization' for instance. This is too complex a phenomenon to be captured by one single discipline.

And this idea of the complexity of the modern world extends to all the big issues of the day: the 'knowledge-based' society and innovation; environmental sustainability; national and personal security; citizenship and identity, for example. None of these is thought to satisfactorily lend itself to a single disciplinary approach. And one could extend these examples to many other contemporary areas of concern.

Gibbons (1999) suggests this is part of a new 'reflexive' contractual settlement between science' and society: "One aspect of this new contract is that it needs to reflect the increasing complexity of modern society. For example, there are no longer clear demarcation lines between university science and industrial science, between basic research, applied research and product development, or even between careers in the academic world and in industry. There is now greater movement across institutional boundaries, a blurring of professional identities and greater diversity of career patterns" (Gibbons, 1999, p. 81).

But there is a second underlying reason why cross-disciplinary approaches are favoured over single disciplinary ones, which has to do with the desire to engage the public in a dialogue with the traditional disciplinary culture and its outputs. The way this is presented, particularly in the UK context, is the demand by funding bodies to engage the constituencies or stakeholders in any research programme. An enormous amount of effort on the part of researchers goes into meeting this requirement for 'user involvement', which has become effectively a binding rule and condition for funding. Such reflexivity creates its own interesting 'imagined' parties in the relationship: the imagined lay person (ILP) (Mareanta, et.al., 2003) and a possible counter in the imagined disciplinary expert (IDE) (Strathern 2004).

However, it goes further than this, since in many cases the demand is to draw the public into the very process of investigation; they effectively become co-researchers in many ways. What seems to be at stake here is the attempt to muddy the boundary between 'experts' and 'lay-opinion'. The implication is that there is no longer a clear distinction to be made between those with expertise and those without it, i.e. the general public. Thus, to some extent at least, the notion of 'expertise' is under threat from the adoption of cross-disciplinary approaches.

Indeed, it implies a complex relationship between expertise and lay opinion in which, perhaps, that distinction itself is being eroded (Strathern 2004). In the UK context this has to do with a more general political programme of 'social inclusion' and the 'democratization' of areas of social and economic life. All research has to be relevant, and a way to ensure this is to include the general public in the process of agenda setting and 'assessing' that research as it goes along. This is made more amenable by pressing cross-disciplinary type approaches since these are not only thought to be more appropriate in dealing with a complex world, but also are more easily understood by the lay-public, enabling them to intervene. In the light of this Gibbons (1999, C.84) suggests that 'reliable knowledge' is being replaced by a new category: 'socially robust knowledge'.

But what might be the consequences of this? The difficulty is that it could simply end up promoting the idea that anyone's opinion is as good as anyone else's. Thus knowledge becomes merely a matter of 'opinion' without any clear way of discriminating between such opinions. And this could equally well provide further justification for 'popular prejudice' to become legitimized. One only has to look to the USA where the ideas about Darwinism are incendiary in popular debate, and where scientists like Richard Dawkins are reduced to shouting at evangelical Christians, and vice versa. But these kinds of disputes are accelerating, and this is not unconnected to the way scientific research is being recast along lines that undermines the distinction between expertise and lay opinion. And this raises an important further connected point. Disputation amongst scientists over the validity of research results is increasing and the scepticism amongst the public similarly growing. These two are inter-linked. Take 'global warming' for instance. Climate scientists cannot agree on whether this is happening, or what or who is responsible if it is. There is widespread publicly voiced disagreement. Similarly with vaccinations like MMR for rubella. Medical scientists disagree on the consequences of this. And there are many other profound disagreements along similar lines. What does the public do under these circumstances where science disagrees? The decision is thrown on to the public (lay opinion) to makes up its own mind, or chose what to do.

Science (expertise) cannot solve the issue one way or another. So the public is asked to assess what it should do 'on-its-own' as it were. It is 'abandoned' to its own devices in making a judgement. How might it go about this? It must detect partisanship and biases amongst experts. Presumably, it does this by assessing the claims and counterclaims of different groups of scientists by looking at their credentials: what universities or research institutes are they from (prestigious/non-prestigious); what is their status within their professional bodies; who publicly backs this or that group; reputation and authority; past record.

And this emphasis on bringing the general public closer into the research agenda setting process is bolstered by another requirement associated with cross-disciplinary: the disciplines and research must be made more 'accountable', as mentioned above. Thus cross-disciplinary is also associated with a desire to make academic and other disciplinary bound bodies more accountable, to both their constituencies of interest and the funding bodies. Cross-disciplinary enhances this process since it is more accessible to funding masters, and less tied up with hidebound single discipline constraints.

An added incentive is provided by the kind of student and their interests that are coming into the higher education system, and the kinds of demands being made by the labour market as they exit that system. Increasingly secondary schools are promoting cross-disciplinary type activity within the curriculum and project work. This is partly to foster student interest and as a strategy for social inclusion, and to deal with mixed abilities. But when they get to universities, their expectations and choices are to build on this with more and more cross-disciplinary type activity. Combined degree and vocationally orientated degree programmes have, as a result, proliferated. And employers want a compliant and adaptable set of work skills, which emerge more readily from cross-disciplinary activity.

Finally, it is worth considering whether cross-disciplinary actually represents a new form of governance, whether in a rather paradoxical manner it represents a new way to (re)-discipline the disciplines? This could appear paradoxical since, ostensibly at least, cross-disciplinary seems to undermine the disciplines, to disturb their hold on academic and research life. But it may indirectly provide the means to gain more access to the disciplines themselves. If the traditional disciplinary structures are seen as one of the main obstacles to 'sensible' and 'relevant' research, one 'fit for the modern world', then promoting cross-disciplinary - in part at least - would provide an avenue to gain added indirect leverage on these disciplines by by-passing them, as it were, or pressing them into the service of cross-disciplinary.

From:
Thompson, G. (2010). Disciplinarity, Cross-Disciplinarity and 'Performance'. Department of Business and Politics. Copenhagen Business School. Working Paper / Department of Business and Politics No. 70

References:

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Strathern, M. (2004) 'Laudable aims and problematic consequences, or: the "flow" of knowledge is not neutral' Economy and Society, Vol.33, No.4, November, pp.550-61.

1.2 READING CRITICALLY II
(Synchronous Tutorial)

In this unit, we will apply what we have learnt about critical reading by evaluating arguments in texts across disciplines. We will also examine more closely how critical attitudes are textualized in different academic texts.

Task 3: Reading

*Please complete **before** attending tutorial*

Read Shehzad (2008), "Move two: Establishing a niche." [Found in Canvas> Course Readings]

As you read take some brief notes on points you find interesting. Try to be a critical reader:

- What key ideas or insights stand out to you?
- Consider what the author leaves space for further exploration. What remains undeveloped or unresolved?

Bring your notes and article to class.

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Task 4: Comparing Warranting in Academic Texts

In academic writing, how authors support their claims—what is often called *warranting*—can vary greatly depending on the discipline and the nature of the argument.

Texts A, B, and C below are extracts from journal articles in the social sciences, natural science, and engineering respectively.

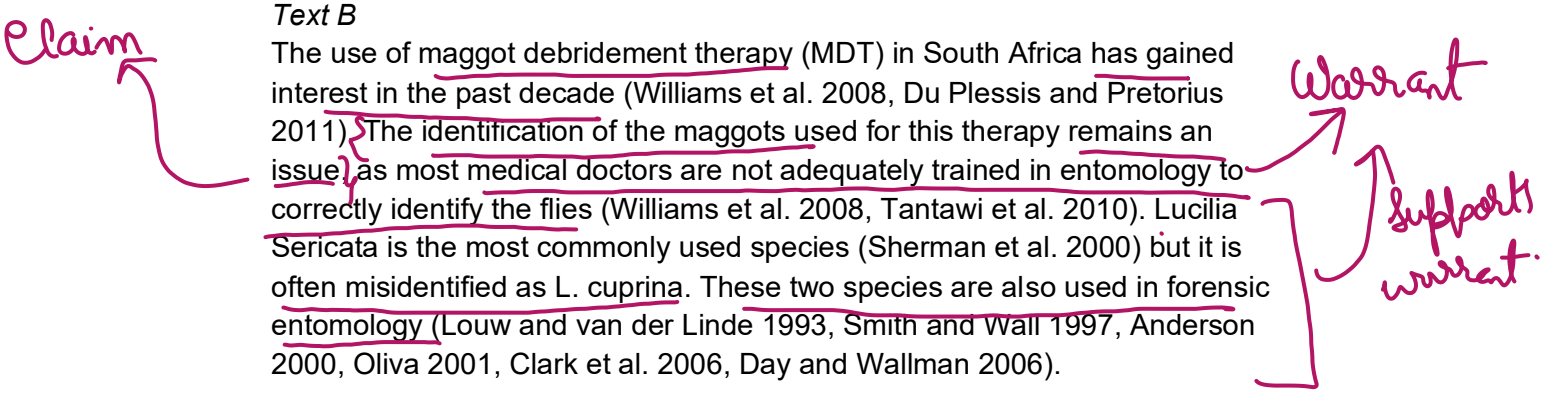
- Compare the *warranting* in their arguments.
- What differences do you observe? Why do you think there are differences?

Text A

Traditional methods of teaching in most school classrooms involve the spoken word as the primary mode of communication. On average, up to 60% of classroom learning activities involve either listening or participating in verbal communication with the teacher or other learners (Sutherland & Lubman, 2001:2) which means the better the learner can hear, the more s/he is able to learn (Smith, 2002:2).

Text B

The use of maggot debridement therapy (MDT) in South Africa has gained interest in the past decade (Williams et al. 2008, Du Plessis and Pretorius 2011). The identification of the maggots used for this therapy remains an issue as most medical doctors are not adequately trained in entomology to correctly identify the flies (Williams et al. 2008, Tantawi et al. 2010). Lucilia Sericata is the most commonly used species (Sherman et al. 2000) but it is often misidentified as L. cuprina. These two species are also used in forensic entomology (Louw and van der Linde 1993, Smith and Wall 1997, Anderson 2000, Oliva 2001, Clark et al. 2006, Day and Wallman 2006).



Text C

Accurate determination of soil resistivity in the local environment is critical to minimize bias in electrical grounding system investigations. Incorrect soil resistivity values can lead to faulty grounding system designs flaws, compromising both safety and functionality. According to the criteria set by the Electricity Generating Authority of Thailand (EGAT), the resistivity of backfill soils in substations must not exceed 80 Ω-m. However, it has been observed that the resistivity of the soil often fails to meet these stringent requirements. This shortcoming necessitates the development of robust methods for assessing soil resistivity.

What makes a literature review critical?

The literature review is an overview of existing scholarship which help you establish, firstly, what the current theoretical or policy debates related to your topic are, and, secondly, what the current state of knowledge related to these issues and problems is.

In the broader academic context, a critical literature review helps you to justify the purpose and significance of your research project to committees and funding sources. By strategically reviewing the published literature on the topic you can show how your proposed research will contribute to the scholarly conversation or impact wider society.

Critical literature reviews are personal. They reflect the intellect of the reviewer, who has decided the focus, selected texts for review, engaged critically with and interpreted the evidence they offer, ordered and synthesized what was found, and written the final account.

Wallace and Wray (2021) define a critical literature review as "a reviewer's constructively critical account, developing an argument designed to convince a particular audience about what the published – and possibly also unpublished – literature (theory, research, practice or policy) indicates is and is not known about one or more questions that the reviewer has framed" (p.179).

Note that the above definition emphasizes that critical literature reviews are designed to persuade. Literature reviews across disciplines should not merely summarize texts or previous studies. Such reviews simply restate what is in the texts rather than **evaluating** them, building an argument, targeting an identifiable audience, or addressing a specific question.

This does not mean that the literature review should be too dismissive or destructively critical. A critical literature review should not indulge in negative evaluation for the sake of demonstrating the authors' foolishness and the reviewer's intellectual superiority.

Critical literature reviews have several features. A critical literature review is likely to be:

- *Focused* on an explicit substantive (i.e. some aspect of the social world), theoretical or methodological review concern;
- *Structured* so as to address each review focus in a logical sequence;
- *Discerning*, so that some texts are given a more in-depth consideration than others, according to the reviewer's judgement of their centrality to the review foci and interpretation of the evidence they offer;

- *Constructively critical*, evaluating the extent to which knowledge claims and the arguments they support are convincing, or whether a theoretical orientation is coherent;
- *Accurately referenced*, so that each source can be followed up by readers of the review;
- *Clearly expressed and reader-friendly*, with interim conclusions and signposting, to help readers get the reviewer's message easily and follow the development of the argument;
- *Informative*, providing synthesis through a strong conclusion which summarizes the reviewer's judgement about how the cited literature answers the review focus, indicating the strengths and weaknesses of the evidence, and arbitrating between any opposing positions reviewed;
- *Convincingly argued*, expressing the reviewer's 'voice' authoritatively because the conclusion is adequately warranted by evidence, as interpreted by the reviewer, drawn from the literature or the reviewer's experience; and
- *Balanced*, indicating that the various viewpoints expressed in the literature have been carefully weighed and that the reviewer's judgements are demonstrably based on a careful assessment of the relevant strengths and limitations of the evidence presented in that literature.

Task 5: Evaluating a Literature Review

Study the excerpt below, which is taken from a section of the literature review draft written by a doctoral student for his research proposal. How critical do you find his literature review?

amateur.

Further discussion:

How effectively does the literature review establish and frame the research question/ research purpose? To what extent does the writer assert their own critical or interpretive perspective?

[1] There are many studies dealing with the effect of KM (knowledge management) on operating performance in modern organizations. Liu, Chen, and Tsai (2004) claimed that knowledge has become a main part of manufacturing resources and a prerequisite for success in the production environment. Their study looked at the relationship between KM capability and competitiveness in Taiwan's industries. They set the functions of knowledge obtaining, knowledge refining, knowledge storing, and knowledge sharing as the measurement factors (independent variables). On the other hand, they looked at the enterprise characteristics, technology advantages, and scale of the enterprise as the dependent variables. Finally, they claimed that KM capability has a tremendous effect on competitiveness.

[2] Liu and Tsai (2007) utilized the balanced scorecard approach to investigate the effects of the introduction of a KM system on the operating performance in Taiwanese high technology (hi-tech) companies. The operating performance includes financial performance, business performance, organization performance, and the long-term resource advantage. They claimed there was a 5% to 10% improvement in performance in the areas of customer, finance, and internal business process, after introducing KM. They concluded that KM had a positive effect on operating performance among the hi-tech firms in their study.

[3] On the other hand, Choi and Lee (2002) did an empirical investigation of KM styles and their effect on corporate performance. They categorized KM methods into four styles. These are the dynamic, system-oriented, human-oriented, and passive styles. They claimed that the dynamic KM style resulted in a higher corporate performance than other styles.

[4] Meanwhile, there are articles written on the impact of KM on Customer Relationship Management (CRM). Bueren et al. (2005) illustrated the use of modern information technology to provide knowledge support to CRM processes and claimed that knowledge support could be used for performance enhancement in customer-oriented business processes. It helps to improve the customer relationship, and hence improve the sales performance of a firm.

[5] There are articles written on the KM application in New Product Development (NPD). Chen et al. (2008) claimed that appropriate KM methods can aid the selection of the right set of New Product Development (NPD) which is critical to a company's long-term success. In fact, resources are not immense, and therefore we cannot develop all new products and have to select the NPD mix strategically. They claimed that a firm could operate innovatively, effectively and efficiently on the NPD project processes if a suitable KM method is included in analyzing the characteristics of a strategic NPD mix. Moreover, KM has always been significant for organizations whose economic success derived from the technological advantage achieved through NPD (Pitt & MacVaugh, 2008). Organizations will rely more on KM in a dynamic technological environment and KM can help to improve product design and eventually the new product development capability.

[6] By the way, there are articles exploring the impact of KM on the delivery of quality. Stewart and Waddell (2008) examined the link between KM and quality management with a focus on the role of a quality culture. They claimed that in order to survive in such a dynamic environment, organizations have to embrace KM as a fundamental component of the delivery of a quality culture. In other words, organizations that are

aware and responsive to the link between KM and quality management will be in a better position to deliver quality to their customers.

[7] Consequently, I have described different literatures on KM application and its effect on the performance of firms.

[8] However, there is not much research that deals with KM directly in the manufacturing firms in China where the institutional arrangements and market conditions are distinct from the developed countries.

[9] In order to fill the gap, my research focuses on how the capabilities of KM can be used in the manufacturing firms operating in China. I will explore how KM affects the areas of sales, quality control, training, inventory control, financial control, customer service, and so on.

Task 6: Literature reviews and authorial voice

In today's academic environment, generative AI tools are increasingly used to help students read and synthesize complex texts. It is important to reflect on how one's own critical reading (which reflects one's own unique point of view and inclinations) compares with AI-generated summaries.

**Compare your reading with an AI-generated summary:*

- In what ways does the genAI summary compare to your own understanding of the research?
- How well do you think genAI can identify problems, limitations, or gaps in the research?
- How might individual creativity or "reading between the lines" shape your interpretation in ways that genAI might not capture?

Reading Critically to Establish a Niche

Swales and Feak (2012) argue that research papers typically compete with other papers for acceptance and high readership in "highly competitive environment[s]" (p. 284). As a result, authors need to offer **thought-provoking questions and original or unresolved problems** (Ahltrom, 2017), such that their research **does not become mere repetition of already existing knowledge** and to distinguish their work from others in the field (Neuman, 2007; Swales & Feak, 2012).

This starts with reading critically: going beyond mere summary to question, evaluate, and synthesize works in ways that reflect one's own unique and idiosyncratic perspective. This sort of engaged reading feeds directly into later stages of writing. It **enables authors to craft literature reviews**—whether as a full chapter or a shorter section in the introduction—**that strategically and persuasively position their research in relation to existing scholarship**. Using various rhetorical and linguistic strategies, scholars can demonstrate how their research problem or question enters into critical dialogue with prior work, by building on, responding to, or challenging it. The way scholars define their research territory is a process also known as *establishing a niche*.

Some ways scholars establish a research niche:

- Address a gap in previous research
- Challenge dominant perspectives or existing assumptions
- Raise new questions
- Focus on overlooked issues or groups
- Extend or build on prior work in a new context or with a new method
- Replicate previous research in a new context
- Correct previous research (with a counterclaim, for example)
- Synthesize across disciplines to generate new insights
- Propose a new theory, concept, model, approach
- Refine or redefine terms, definitions, categories
- Revise understanding through new or updated data
- Respond to new, emerging, or critical societal needs

Scholars emphasize the importance of their niche by

- Highlighting real-world importance or relevance (e.g. social, political, practical)
- Highlighting scholarly urgency (e.g. inconsistencies, uncertainties, unresolved tensions in scholarship)
- Framing their work as new or timely
- Emphasizing practical utility
- Highlighting limitations in existing research and offering a solution
- Connecting to broader disciplinary conversations

Task 7: How scholars establish their niche

Read the following peer-reviewed article introductions (Texts A-C). How do the authors establish their niche? Pay close attention to how the literature review contained within helps to position the research within a given field or topic. Also pay attention to how their *language* helps to establish the importance of their niche.

Text A

There are an estimated 1.2–5.5 million snakebites annually around the world, with approximately 20,000–94,000 attributed annual deaths [1]. The annual cost burden of snakebites is significant, and worldwide morbidity and mortality represent a neglected global public health problem [1,2]. In the United States, there are approximately 7,000–10,000 annual emergency department visits from snakebites [3], with 32–60% from venomous species, the majority being pit vipers [3,4], and 5–7 annual fatalities per year [3-5].

Snakes are poikilothermic animals whose activities, life cycle, and geographic distribution are closely tied to ambient temperature [7,8], and vary as a result of climate change [9]. There is at minimum a 99% chance that average global temperatures have significantly increased since 1950 to present times [11], and this global warming or "climate energizing" frequently results in increased extreme weather patterns [12,13].

These fluctuations of extreme weather (both drought and high precipitation) impact human health through an increase of natural disasters such as intense heat waves, droughts, and coastal flooding [14]. Likewise, climate change in North America is associated with changing distribution of venomous species that may lead to increased human morbidity [12,15,16]. Climate change has been described as the greatest global threat of the twenty-first century [17], but the effect on snakebites in North America has not been examined.

There is some evidence for increased incidence of snakebites with warming temperatures, however interpretation is limited to Central American meteorological patterns [9], and the current literature on snakebites in the United States provide limited insight into this relationship [3,18]. With the recent most severe drought and precipitation seasons ever recorded in California [19], we sought to correlate the relationship of climate trends with snakebite incidence in California over 20 years. The study's primary hypothesis was that the severity of drought would correlate with increased incidence of snakebites and could be predicted by weather patterns.

Source: Adapted from Phillips C., Lipman G., Gugelmann H, Doering K., & Lung D., "Snakebites and climate change in California, 1997-2017," Clinical Toxicology, 2019.

Text B

Intro

Indigenous peoples make up over 5% of the world's population, their territories comprise at least 20% of the land area of the planet, and these lands host an estimated 80% of the world's biodiversity (Stevens 2014). The relationships between indigenous land tenure and biological diversity is a challenging area of research, but several studies suggest that indigenous peoples' place-based values, institutions, and practices help promote biodiversity.

Lit gap

Indigenous lands often support higher native and rare species richness (Redford and Stearman 1993, Peres 1994, Arcese et al. 2014) and lower rates of deforestation (Nolte et al. 2013, Ceddia et al. 2015). However, beyond a handful of studies looking at biodiversity and land use change, few researchers have looked at the broader ecological outcomes of indigenous land tenure.

Example

Indigenous peoples in Wisconsin manage forestlands and wildlife by merging professional standards of forestry and wildlife practice with their own culturally specific traditional ecological knowledge and land-based values. Although commercial and ecological forestry best-practices are utilized in tribal forestry programs, indigenous communities' own knowledge systems and values have a strong influence. This has often led to distinct forest and wildlife management goals, institutions, and practices (Trosper 2007, Dockry 2012, Reo and Whyte 2012). Indigenous forestry in the United States and Canada has several unique aspects (Bengston 2004). The Cherokee Nation of Oklahoma, for example, plants a diverse suite of culturally significant tree species and favors nonchemical methods of land clearing that promote the growth of food and medicinal plants while protecting harvesters' health (Carroll 2015). Anishnaabe of Shoal Lake in Northwestern Ontario use succession management via landscape burning and timber harvest to promote biodiversity and support local livelihoods (Berkes and Davidson-Hunt 2006).

Indigenous Nations and nonindigenous forest and wildlife managers generally profess similar goals typically centered around sustaining resource values and yields, biological diversity, and ecosystem productivity despite some differences. Do these shared goals lead to similar outcomes on the lands managed by these two groups? Here, we explore that question within the specific context of northern Wisconsin where a mix of private industrial, private nonindustrial, and county, state, and national public forestlands surround four major "Indian reservations," i.e., lands under jurisdiction of Indigenous Nations in the U.S. (Fig. 1A).

Specific area

This study thus fits within existing literature concerning the foundations and outcomes of indigenous forest and wildlife management. Our focus here, however, is to quantify the ecological outcomes of forest and wildlife management and links between the two. That is, have the land-based values and practices of indigenous peoples in Wisconsin led to significantly different ecological conditions on their forestlands relative to neighboring lands? In the discussion, we return to the question of how differences in values and goals contribute to the differences in outcomes we document.

Source: Adapted from Waller, D., and Reo, N., "First stewards: ecological outcomes of forest and wildlife stewardship by indigenous peoples of Wisconsin, USA," Ecology and Society, 2018.

Text C

For wind energy to become competitive with respect to other sources of energy, the initial consideration must be to reduce the cost of energy from wind power. In modern wind power research, how to minimize the cost of a wind turbine per unit of energy is an important task. The shape of the rotor blades plays a decisive role in determining the overall aerodynamic performance of a horizontal axis wind turbine (Fuglsang, 1995; Giguere and others, 1999; Benini, 2002). Thus, aerodynamic optimization of the blade shape is a very important stage in the design and manufacturing of wind turbines.

In earlier days, the Glauert and Wilson methods were mostly used for blade design (Spera, 1994). The objectives of these methods were to obtain the maximum power coefficient of each blade section at the design wind speed. Because the time variation characteristics of wind speed are not taken into account, blades designed by these methods cannot achieve the maximum annual energy output. Furthermore, design results from these methods must be substantially corrected to get smooth chord length and twist distributions (Spera, 1994). Because corrected results already deviate from the design points, effectively controlling design results poses problems. Conventional search algorithms, such as the feasible direction method and the complex method, are often prone to converging on the local optimal point (Gen, 1999). For some complicated problems, it is difficult to obtain a global optimal result.

Several studies have been carried out on the viability of the wind turbine for power generation. Grauers, (2003), worked on efficiency of three wind energy generator systems. Jureczko and others, (2005), developed optimization of wind turbine blades. Afolami, (2007), improved the operation of a wind turbine through replacement of some components of the wind turbine. Michael, (2007), showed that modern wind turbines have become an economically competitive form of clean and renewable power generation. In his study, he carried out an economic optimization analysis of a variable speed, three blade, and horizontal-axis wind turbine.

This work presents design tool for optimizing wind turbine blades. This considers optimizing the blade of wind turbines with respect to maximizing the energy yield of a wind turbine. To achieve the optimization in this work, the genetic algorithm code in MATLAB is written for the turbines and this code can be used for any type of wind turbine provided that the objective function is the same.

Source: Adapted from Eke, G. B., & Onyewudiala, J. I. (2010). *Optimization of wind turbine blades using genetic algorithm*. Global Journal of Researches in Engineering: Mechanical and Mechanics Engineering, 10(7), 20–29.

Follow-up activity:

Search for a leading journal in your field, then use relevant keywords to find three research articles related to your interests. Review their introductions, focusing on how the authors position their work in relation to prior studies. How do their approaches compare to the examples above?

In summary, reading critically means being discerning about the warranting of conclusions and making informed judgments about the quality of arguments by synthesising information gleaned from asking critical questions of academic texts.

Critical reading is an essential skill for graduate researchers as it ultimately helps them read the field or discipline critically, asking critical questions such as:

- On what basis was the field established?
- Who got to speak? who gets to speak?
- What was written about what, when and for whom? How has this changed over time and in what ways?
- In whose interests did this research and writing work? Does it still work this way? How is it changing?
- What kind of knowledges, interpretations and authors are missing or marginalised?
- Are there any patterns to these omissions and sidelining?

Such a critical reading would create new possibilities for their project, and ultimately help them to establish their own research niche, while at the same time contributing towards producing a more equitable field or discipline.

***Task 8: Required Supplementary Unit**
Independent e-learning task

The Required Supplementary Unit opens in Week 2 and closes in Recess Week.

This unit comprises a set of three e-modulets on special topics on academic communication for graduate researchers.

Each e-modulet further comprises:

- an e-lecture (video) and
- a quiz based on that lecture.

Please watch the e-modulet (video) before attempting the quiz.

You are required to *complete this supplementary unit, at your own time and pace*, before the deadline (**Sunday 28 September, 23:59**).

The e-lectures may be found in Canvas Modules > Required Supplementary Unit; the quizzes can be found in Canvas > Quizzes.

- E-Modulet 1 (Required Supplementary Unit): The semantics of logical connectors;
- E-Modulet 2 (Required Supplementary Unit): Writing about methods;
- E-Modulet 3 (Required Supplementary Unit): The results and discussion sections.

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